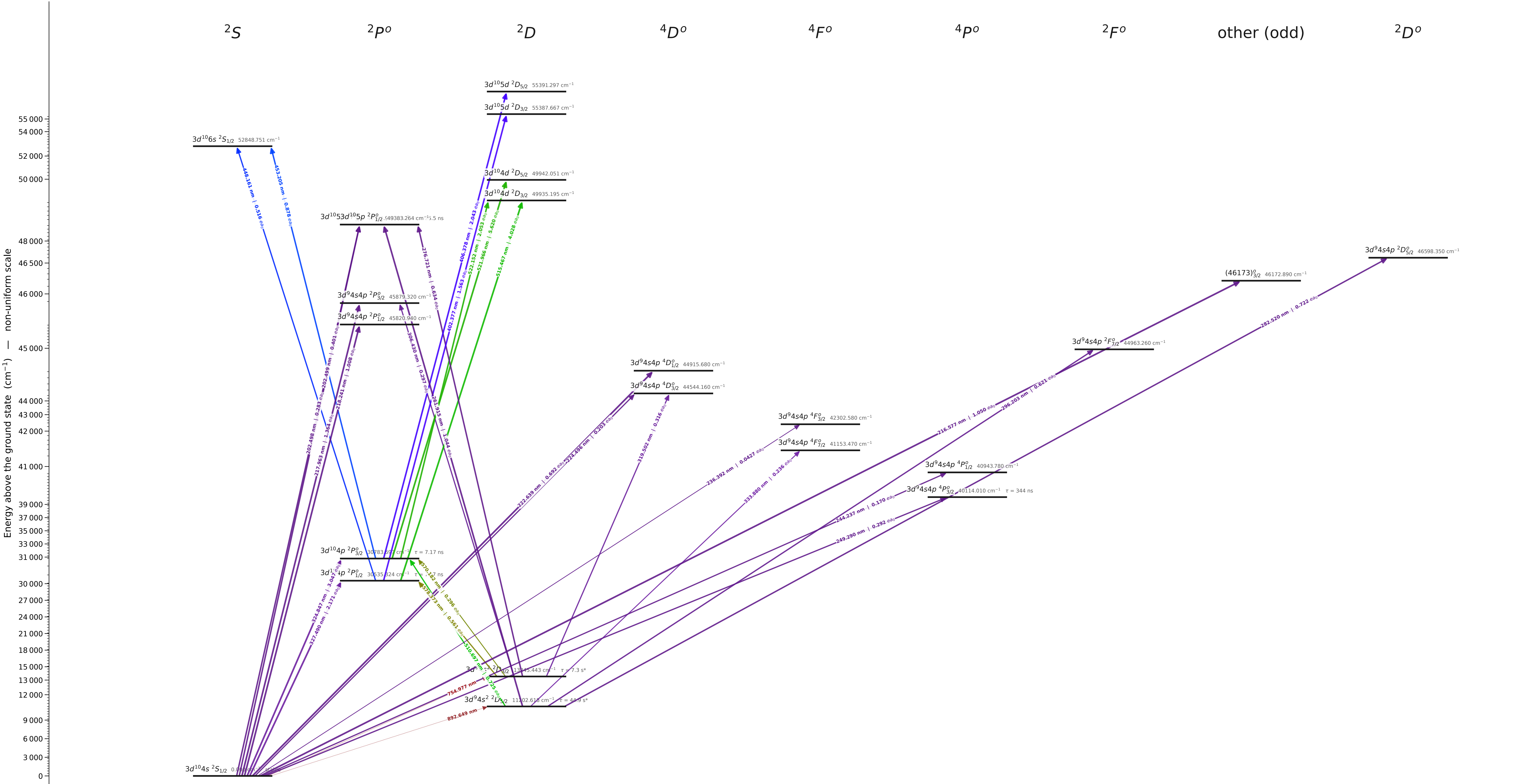
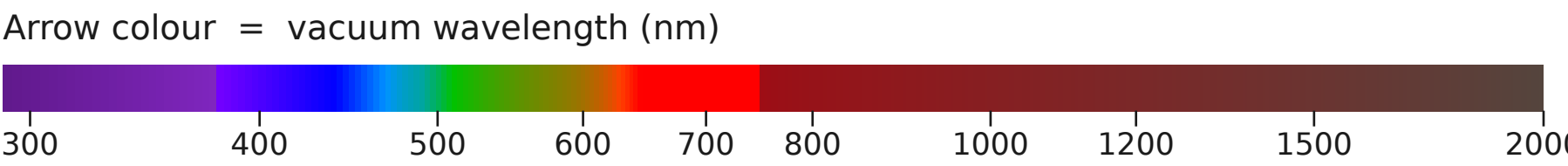


Copper

Cu (Cu I). Strongest classified transitions below 2 μm (NIST): 23 levels, 31 lines.

Cu ⁺ ionisation limit 62317.46 cm⁻¹ = 7.72638 eV (160.469 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||U'|$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	U er U' (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d ¹⁰ 4s ² S _{1/2} - 3d ¹⁰ 5p ² P _{3/2}	202.4991	202.4339	1480.4634	0.401	9.800e+06	6.241	25	202.4 nm line used for pulsed level-crossing spectroscopy...
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s4p ⁴ P _{1/2}	244.2373	244.1633	1227.4636	0.170	2.010e+06	-	-	244.2 nm line used for in-gas-cell resonance-ionisation s...
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s4p ⁴ P _{3/2}	249.2895	249.2143	1202.5878	0.292	2.790e+06	0.4627	96	249.2 nm line used for resonance-ionisation spectroscopy...
3d ¹⁰ 4s ² S _{1/2} - 3d ¹⁰ 4p ² P _{3/2}	324.8473	324.7536	922.8720	3.047	1.372e+08	22.2	98.4	D2 resonance line 324.8 nm
3d ¹⁰ 4s ² S _{1/2} - 3d ¹⁰ 4p ² P _{1/2}	327.4896	327.3952	915.4260	2.171	1.359e+08	21.89	98.8	D1 resonance line 327.4 nm
3d ⁹ 4s² ² D _{5/2} - 3d ¹⁰ 4p ² P _{3/2}	510.6971	510.5548	587.0260	0.725	2.000e+06	22.2	1.43	510.6 nm green copper-vapour-laser line
3d ⁹ 4s² ² D _{3/2} - 3d ¹⁰ 4p ² P _{1/2}	578.3730	578.2126	518.3376	0.561	1.650e+06	21.89	1.2	578.2 nm yellow copper-vapour-laser line
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s² ² D _{3/2}	754.9766	754.7688	397.0884	-	6.270e-07	2.18e-08	0	second proposed clock transition
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s² ² D _{5/2}	892.6485	892.4035	335.8460	-	2.230e-02	3.545e-09	100	E2 clock transition proposed for a Cu lattice clock

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁶³Cu hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
3d ¹⁰ 4s ² S _{1/2}	5866.915(5)	-	F=1: -7333.64 F=2: +4400.19
3d ⁹ 4s² ² D _{5/2}	749.100	186.000	F=1: -3802.57 F=2: -2453.18 F=3: -289.57 F=4: +2855.62
3d ⁹ 4s² ² D _{3/2}	1851.000	137.400	F=0: -6769.50 F=1: -5055.90 F=2: -1491.30 F=3: +4199.10
3d ¹⁰ 4p ² P _{3/2}	506.600	-	F=1: -633.25 F=2: +379.95
3d ¹⁰ 4p ² P _{1/2}	194.5(11)	-28.0(6)	F=0: -764.38 F=1: -541.88 F=2: -124.88 F=3: +430.62
3d ⁹ 4s4p ⁴ P _{3/2}	2129.000	-	F=0: -7983.75 F=1: -5854.75 F=2: -1596.75 F=3: +4790.25
3d ⁹ 4s4p ⁴ P _{1/2}	2432(8)	-	F=1: -3040.00 F=2: +1824.00
3d ⁹ 4s4p ⁴ F _{7/2}	1139.000	-	F=2: -7688.25 F=3: -4271.25 F=4: +284.75 F=5: +5979.75
3d ⁹ 4s4p ⁴ F _{5/2}	305.000	98.000	F=0: -1021.25 F=1: -814.25 F=2: -302.25 F=3: +710.75
3d ⁹ 4s4p ⁴ D _{3/2}	124.000	-3.000	F=0: -468.75 F=1: -341.75 F=2: -90.75 F=3: +278.25
3d ⁹ 4s4p ⁴ D _{1/2}	-510.000	-	F=1: +637.50 F=2: -382.50
3d ⁹ 4s4p ² F _{7/2}	1919.000	-	F=2: -12953.25 F=3: -7196.25 F=4: +479.75 F=5: +10074.75
3d ⁹ 4s4p ² P _{3/2}	948.000	-	F=1: -1185.00 F=2: +711.00
3d ⁹ 4s4p ² P _{1/2}	1619.000	27.000	F=0: -6037.50 F=1: -4445.50 F=2: -1234.50 F=3: +3649.50
(46173) _{3/2}	2100.000	-	F=0: -7875.00 F=1: -5775.00 F=2: -1575.00 F=3: +4725.00
3d ⁹ 4s4p ² D _{5/2}	2533.000	-	F=1: -13298.25 F=2: -8232.25 F=3: -633.25 F=4: +9498.75
3d ¹⁰ 5p ² P _{3/2}	61.7(9)	4.9(7)	F=0: -225.25 F=1: -168.45 F=2: -49.95 F=3: +140.05
3d ¹⁰ 5p ² P _{1/2}	334.600	-	F=1: -418.25 F=2: +250.95
3d ¹⁰ 4d ² D _{3/2}	67.000	-	F=0: -251.25 F=1: -184.25 F=2: -50.25 F=3: +150.75
3d ¹⁰ 4d ² D _{5/2}	111.000	-	F=1: -582.75 F=2: -360.75 F=3: -27.75 F=4: +416.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3d ¹⁰ 4s ² S _{1/2} - 3d ⁹ 4s4p ⁴ P _{3/2}	249,290	63-65	1055.4(26)
3d ⁹ 4s² ² D _{5/2} - 3d ¹⁰ 4p ² P _{3/2}	510.697	63-65	-2128.00
3d ¹⁰ 4p ² P _{1/2} - 3d ¹⁰ 4d ² D _{3/2}	515.467	63-65	363.00
3d ¹⁰ 4p ² P _{3/2} - 3d ¹⁰ 4d ² D _{5/2}	521.966	63-65	354.00

Shift = ν(first isotope) - ν(second isotope).